

# Questions for Tomorrow

WATR Digital Twin · July 8, 2026 · Prepared by: Rohel

## A. Sensor Team — Soil Moisture & Calibration Questions

### A.1 UNDERSTANDING SENSOR DRIFT

**Q1.** When a soil moisture sensor drifts, does it typically read too HIGH or too LOW over time? Does the direction depend on the sensor type or the environment it is placed in?

→ We assumed positive drift (reads too high) in our model — is this the common case for your sensors?

**Q2.** If the sensor drifts positively (reads too high), the apparent daily moisture drop looks smaller than the true drop — the pot seems to lose water more slowly than it actually does. Is that understanding correct?

→ Example: true drop is 3 mm/day, but drifting sensor shows only 1.5 mm/day.

**Q3.** What are the typical physical causes of drift in your sensors? Is it age, temperature, soil composition, electrical interference, or something else?

**Q4.** How quickly does drift typically build up — over days, weeks, or months?

### A.2 CALIBRATION BOUNDARY — WHO CORRECTS WHAT?

**Q5.** Our soft calibration adjusts  $m$  (the microclimate factor) to match sensor drops to model predictions. If the sensor is drifting, soft calibration pulls  $m$  in the wrong direction — effectively embedding the sensor error into  $m$ . Should soft calibration be paused as soon as you flag a sensor as drifting?

→ This is critical: a corrupted  $m$  leads to wrong  $ET_{crop}$ , which could under-water the plant. soft cali nothing to do with drift

**Q6.** After drift is detected and the sensor is repaired or replaced, does a new hard calibration always need to happen to reset the absolute anchor? Or can soft calibration recover on its own if given enough data? if sensor is replaces then yes. if only drift then possibly no

**Q7.** Is there a way to distinguish — from data alone — whether  $m$  changed because of a genuine microclimate shift versus sensor drift? Or is that ambiguous without an explicit drift flag from your group?

**Q8.** Hard calibration corrects the constant offset at  $t = 0$ . Soft calibration then adjusts  $m$  for microclimate changes. **If soft calibration runs on a drifting sensor, it will start to absorb the drift into  $m$ .** Is there a threshold (number of bad batches, magnitude of  $m$  shift) after which  $m$  should be considered unreliable and flagged for manual review? This part is wrong

**Q9.** If soft calibration lowered  $m$  to, say, 0.5 because of a drifting sensor,  $ET_{crop} = m \times K_c \times ET_0$  becomes half of what it should be. This means the irrigation model thinks the plant needs half as much water as it actually does. Is this a real risk? Has this happened before?

A drifting sensor doesnt change the value of  $m$ ,  $m$  remains constant. But  $m$  does changes when the season changes, and the drift in the sensor doesnt necessarily has to be due to seasonal change

### A.3 DATA INTERFACE

**Q10.** What MQTT topic do you publish live sensor readings on? What is the payload format (JSON? field names?)

**Q11.** What is the time resolution of sensor readings — per minute, per hour, or per day?

- Q12.** Do you publish any quality flag or confidence score alongside each reading? For example, a field that says 'reading may be unreliable'? [they do actually](#)
- Q13.** When you deliver sensor data, is it the raw voltage converted to percentage, or does your pipeline already apply a sensor-specific calibration curve?

## B. Anomaly Detection Group — Questions

### B.1 DETECTION METHOD

**Q1.** What criteria do you use to determine that a sensor is drifting? Is it a statistical test, a fixed threshold, cross-comparison with neighbouring sensors, or something else?

**Q2.** How quickly after drift starts do you typically flag it? What is the typical lag between drift onset and detection?

→ This matters for us because every batch soft calibration runs during that window will produce an incorrect m.

**Q3.** How do you distinguish between a drifting sensor and genuine soil behaviour — for example, a pot that stays very dry for a long time and really does have a small moisture drop each day?

[we dont see and we dont care what is genuine soil behaviour at all. We just know the soil moisture value from the sensor](#)

### B.2 COMMUNICATION PROTOCOL

**Q4.** Do you publish a sensor health flag on MQTT? If yes, what topic, what payload format, and what values does the flag take (e.g. healthy / drifting / failed)? [Ask them for the status of recalibration](#)

**Q5.** When a drift event is resolved and the sensor is confirmed healthy again, do you publish a resolution message? Do we need to subscribe to a separate topic for that?

[If sensor gives off values, how do you correct it? Does it then live with the known offset and drift or you is the offset and drift completely removable.](#)

**Q6.** Should our soft calibration subscribe to your health flag and pause updates automatically, or do we coordinate through a manual process or shared dashboard?

[we only do subscription for hard calibration part not soft](#)

### B.3 RECOVERY AFTER A DRIFT EVENT

**Q7.** If a drift event happened and soft calibration ran during that period (adjusting m incorrectly for several batches), is there a way to know how many days of bad data were used? Can you provide a timestamp for when drift started? [Again, drift doesnt affect the value of m during soft calibration.](#)

→ This would tell us how many batches to discard and whether we need a hard recalibration.

**Q8.** After you resolve a drift event and confirm the sensor is healthy, our policy says to do a hard recalibration before restarting soft calibration. Who is responsible for triggering this — you, us, or a shared manual step? [after p01 does recalibration to the sensor, we should hard calibrate again.](#)

**Q9.** Do you have a record of past drift events for our sensors? If yes, can we access it to understand how often drift has occurred historically?